DNN - Project 2 - Report

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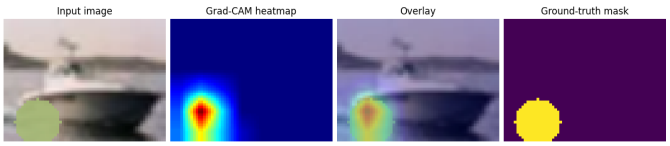
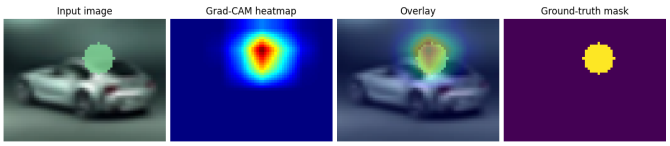
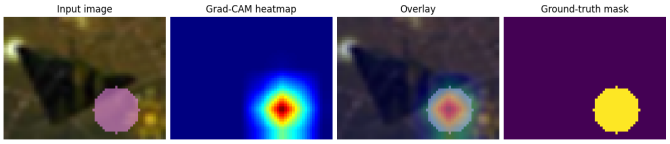
1 Grad-CAM

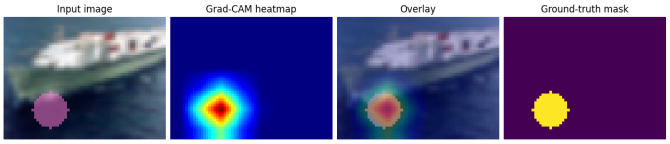
I implemented Grad-CAM as described in the paper. The results are presented below:

1.1 Circles

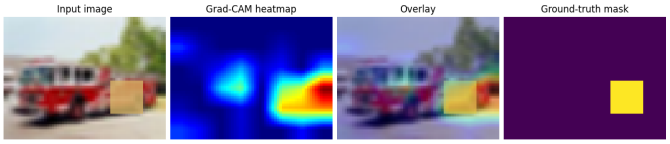


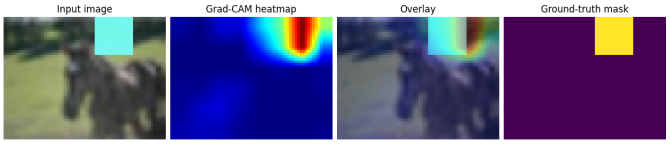


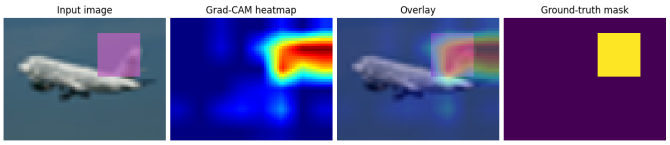


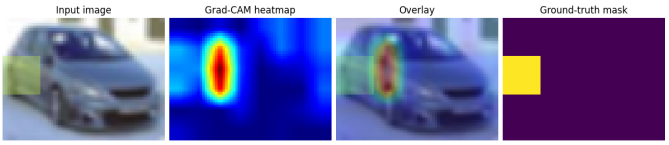


1.2 Squares

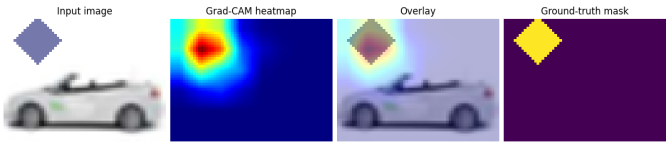


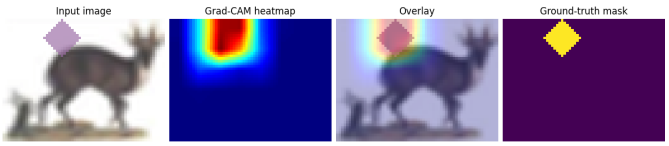


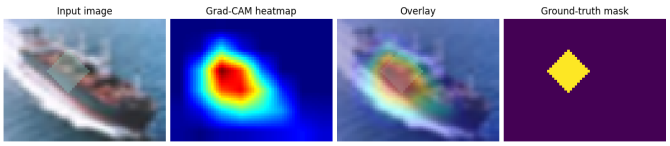


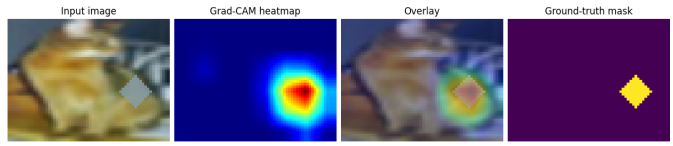


1.3 Diamonds

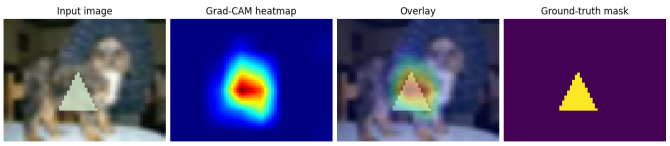


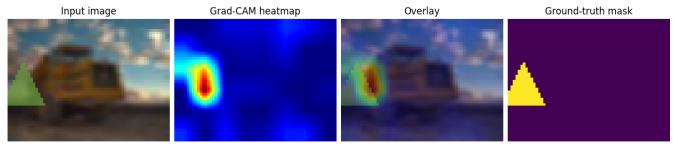


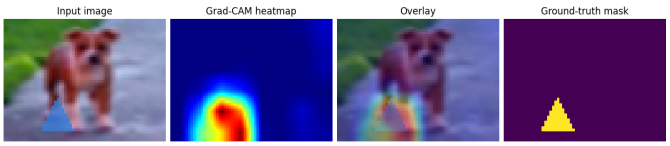


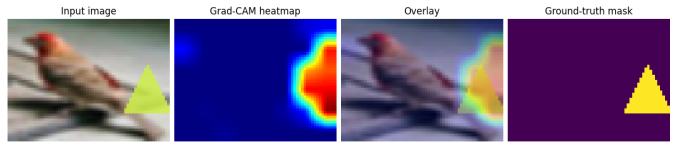


1.4 Triangles

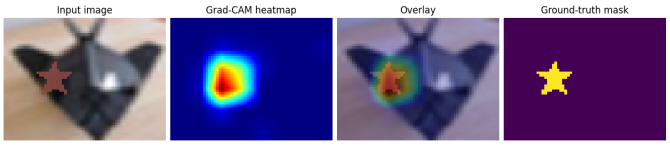


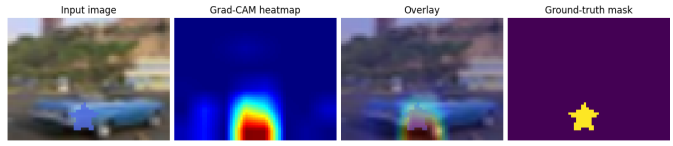


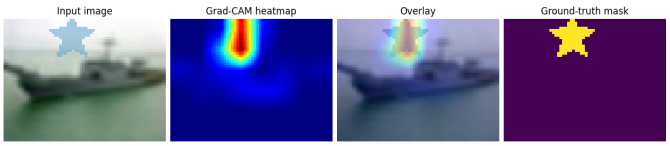


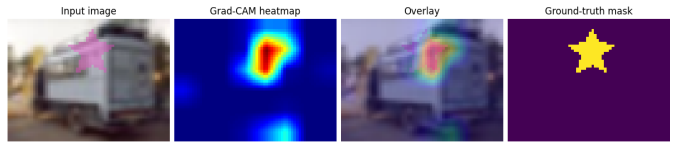


1.5 Stars









The results appear promising regarding point selection for SAM for circles, diamonds, and stars. However, for squares and triangles, the model exhibits a bias towards the right-side edges of the shapes. This observation will be taken into account when selecting foreground points for SAM.

2 SAM

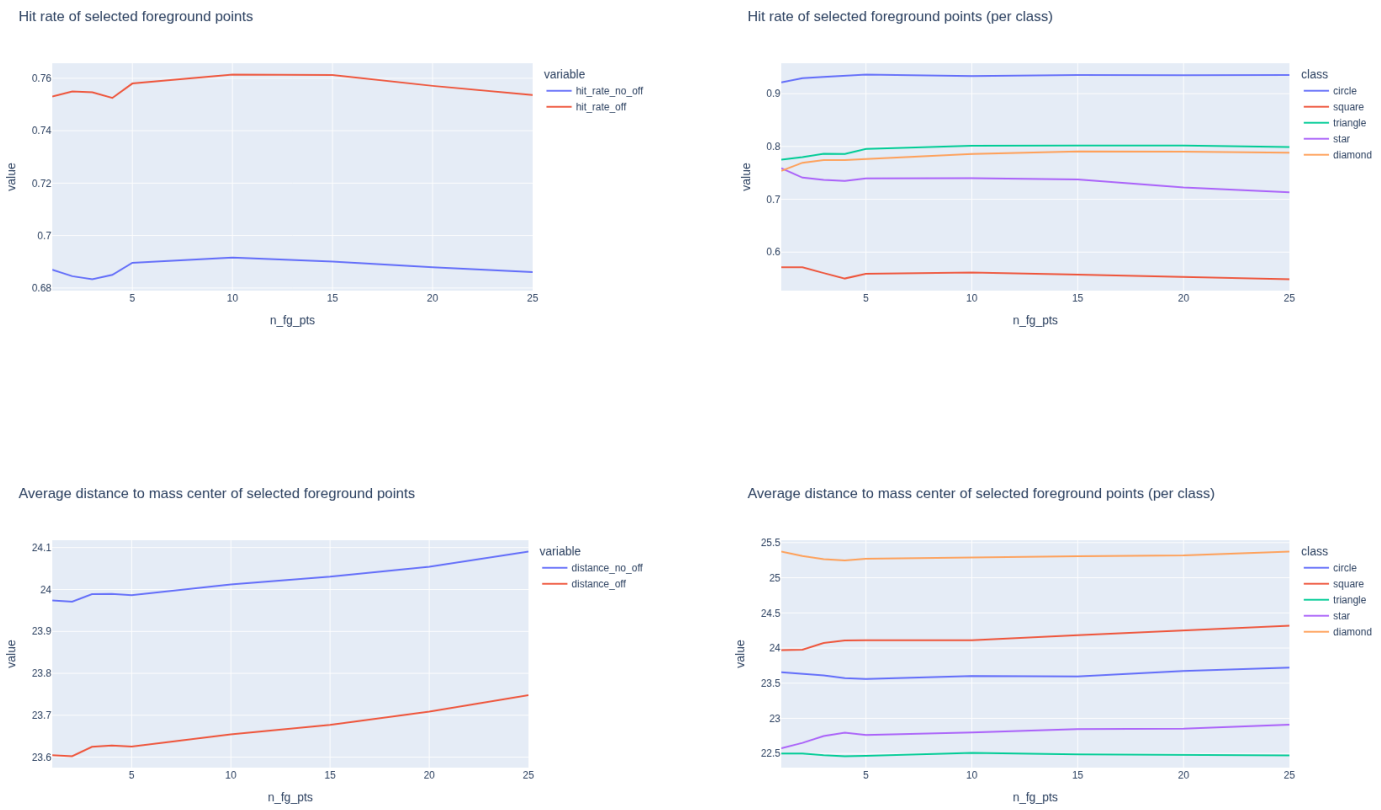
I implemented the following two proposed approaches: 1. At least one foreground point, without background points. 2. At least one foreground point and at least one background point.

2.1 Point selection

Foreground points are selected using the Grad-CAM heatmap by choosing the top n points with the highest heatmap values. Due to the behavior of Grad-CAM on the shapes described above, offsets are applied. Specifically, the distance between the pixel with the highest heat value and the center of the ground-truth mask was computed for each sample in the training set and then averaged.

For background points, the n-th smallest heat value is selected, and 0.001 is added to define a threshold, ensuring that the selected points belong to the background. Among all points with heat values below this threshold, the n points closest to the center of mass of the mask are chosen. The motivation is to select informative background points that SAM can use to separate the foreground from the background.

2.2 Results



The selected approach performs well, and introducing the offset evidently improves the results. Based on the hit rate and distance metrics, 5 foreground points were selected. Additionally, based on experimentation with background points, 5 background points were used.

The results obtained are as follows:

- Hit rate: 0.76
- Distance: 23.60
- IoU for SAM with only foreground points: 0.80
- IoU for SAM with foreground and background points: 0.78

It was observed that, in this task, adding background points is consistently detrimental to the results. This likely comes from the fact that it is relatively easy for SAM to identify the full shape when correct foreground points are provided. Conversely, when foreground selection fails, background points provide little assistance.

The primary method to improve results would likely be a more robust foreground point selection strategy. The current method for determining offsets is simplistic and does not always yield optimal points. For instance, separate methods could be designed for different classes.

Furthermore, applying offsets is very likely to produce poor outcomes when the classifier predicts incorrectly; notably, the provided classifier has an accuracy of 87.4%.